Ref. No. : 385-10-0728

# SAFETY DATA SHEET (SDS)

## 1. Chemical Product and Company Identification

Amilac１０００ Ｂ４２－５０Ｌ semi-gloss

Group Code No. : 371-C
Product Code : ##-371-C00
Product Name : Amilac 1000 color C group
Poisonous and Deleterious Substances Control Law : N.AP.
Type : Aminoalkyd resin coating
Use : Metal coating


Name : KANSAI PAINT CO., LTD.
Address : 6-14, Imabashi 2-chome, Chuo-ku, Osaka 541-8523, Japan
Section in Charge : Tech. Dept. Functional Materials Industrial Coatings Div.
Telephone No. : 03-3732-8115 Fax. No. : 03-3735-3839
Person in Charge : Tech. Dept. Functional Materials Mgr.
Name of Preparer : TAKAHITO YAMADA Url : http://www.kansai.co.jp
Emergency Telephone No. : 03-3732-8115
Night Time Telephone No. : 03-3732-8111

Date Prepared : Oct. 26. 2014

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## 2. Hazards Identification

{GHS Classification}

| | |
|---|---|
| Flammable liquids | Category 2 |
| Acute toxicity(inhalation:gases) | Category 4 |
| Acute toxicity(inhalation:vapours) | Category 4 |
| Skin corrosion/irritation | Category 2 |
| Serious eye damage/eye irritation | Category 2 |
| Carcinogenicity | Category 1A |
| Reproductive toxicity | Category 1B |
| Specific target organ systemic toxicity-Single exposure (Respiratory, Nervous, Liver, Kidney) | Category 1 |
| (Narcotic effects) | Category 3 |
| Specific target organ systemic toxicity-Repeated exposure (Respiratory, Nervous) | Category 1 |
| Acute toxicity to the aquatic environment | Category 2 |
| Chronic toxicity to the aquatic environment | Category 3 |

{GHS label elements}

Danger

{Hazard statements}

- Flammable liquid. May cause explosion when its vapor becomes dense.
- When paint sludge, paint dust and rag soiled by paint oxidize in the air, and generate heat or store heat. They may spontaneously ignit.
- Acute harmful if inhaled.
- May cause organic-solvent toxication.
- May cause skin irritation.
- Cause serious eye irritation.
- May cause cancer.
- Contains substance that may gravely damage health.
- May damage fertility or the unborn child.
- May cause allergic skin reaction.
- May cause respiratory irritation;or May cause drowsiness and dizziness

·Cause damage to organs through prolonged or repeated exposure.
·Toxic to aquatic life.
·Harmful to aquatic life with long lasting effects.

{Precautionary statements}
<Prevention>
· Keep container tightly closed.
· Keep away from ignition sources such as heat/sparks/open flame.
  - No smoking.
· Wear protective gloves and eye/face protection as specified by the competent authority.
· Ground/Bond container and receiving equipment
  - if electrostatically sensitive material is for reloading.
  - if product is as volatile as to generate hazardous atmosphere.
· Use explosion-proof electrical/ventilating/lighting equipment.
· Take precautionary measures against static discharge.
· Use only non-sparking tools.
· Use only outdoors or in a well-ventilated area.
· Avoid breathing dust/fume/gas/mist/vapours/spray.
· Wear protective gloves as specified by the competent authority.
· Wash thoroughly after handling.
· In case of fire:Evacuate area and fight fire remotely due to the risk of explosion.
· Obtain special instructions before use.
· Do not handle until all safety precautions have been read and understood.
· Use personal protective equipment as required.
· Do not eat, drink or smoke when using this product.
· Wash hands thoroughly after handling.
· Do not breathe dust/fume/gas/mist/vapours/spray.
· Avoid release to the environment if this is not the intended use.
<First Aid Measures>
· If water increases risk. In case of fire, use appropriate media specified by the competent authority.
· IF ON SKIN (or hair): Remove/take off immediately all contaminated clothing. Rinse skin with water/shower.
· IF INHALED:Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/physician if you feel unwell.
· IF ON SKIN: Wash with plenty of soap and water.
· Take of contaminated clothing and wash before re-use.
· If skin irritation occurs, seek medical advice/attention.
· Specific treatment reference to supplemental first aid instruction may include a cleansing agent.
· IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
· If eye irritation persists, get medical advice/attention.
· Wash hands after handling.
· IF exposed or concerned: Get medical attention/advice.
· IF exposed: Call a POISON CENTER or doctor/physician.
· Specific treatment reference to supplemental first aid instruction if immediate measures are required.
· Call a POISON CENTER/doctor/physician if you feel unwell.
· IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing.
· If immediate administration of antidote is required.
<Storage>
· Store in cool/well-ventilated place.
· Store locked up.
· Store container tightly closed in well-ventilated place if product is as volatile as to generate hazardous atmosphere.
<Disposal>
· Dispose of contents/container in accordance with local/regional/national/international regulation.

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3. Composition, Information On Ingredients

Component and Concentration of Hazardous Substance

| No. | Components Name | CAS No. | Wt (%) | P R T R | Remark |
|---|---|---|---|---|---|
| 1) | Amorphous silica A | 7631-86-9 | 0.1 - 1 | | |
| 2) | Silane, dichlorodimethyl, reaction product | 68611-44-9 | 0.1 - 1 | | |
| 3) | Titanium Dioxide | 13463-67-7 | 15 - 20 | | |
| 4) | Iron hydroxide oxide | 20344-49-4 | 1 - 5 | | |
| 5) | Xylene | 1330-20-7 | 10 | 1 80 | |
| 6) | Ethyl Benzene | 100-41-4 | 9.6 | 1 53 | |
| 7) | Methanol | 67-56-1 | 0.1 - 1 | | |
| 8) | Isobutyl Alcohol | 78-83-1 | 10 - 15 | | |
| 9) | Formaldehyde | 50-00-0 | 0.41 | S1 411 | |
| 10) | Oxidative-polymerizable Resin | ---- | 25 - 30 | | |
| 11) | C.I.Pigment green 7 | 1328-53-6 | 0.1 - 1 | | |

(Element Name)
None

* In case of the same No. in elements and compounds, use the content of elements for PRTR calculation.

## 4. First Aid Measures

Inhalation
- If disorder occurs for inhalation of vapor or gas, remove to fresh air and keep at rest in a position comfortable for breathing. If feels unwell, call a doctor/physician.
- If disorder occurrs for inhalation of vapor or gas, remove immediately to fresh air and rest with keeping warm. Perform artificial respirationif breathing get irregular or stops. Do not make swallow vomit.

Skin Contact
- Wipe quickly clinging matter with clean cloth.
- Flush with plenty of water and soap. Do not use thinner or solvent.
- Get medical attention, with "SDS," if symptom or pain persists.
- Immediately take off all contaminated clothing.

Eye Contact
- Flush immediately with plenty of clean water for 15 minutes or more.
- Consult physician with"SDS" as soon as possible.
- Immediately call a POISON CENTER or doctor/physician.

Ingestion
- If one swallows accidentally, immediately rest and consult physician with " SDS."
- Avoid forcing to vomit. Do not make swallow vomit.
- Avoid forcing to vomit without physician's instruction.

Protection of a person taking a step temporarily
- Wear proper protectors(safety goggles, respirator, gloves, etc.). Well ventilated.

## 5. Fire Fighting Measures

Extinguishing Media: Carbon dioxide gas Foam Powder

Prohibited extinguisher: Water (cylindrical w

Fire Fighting Instructions
- Wear proper protectors (heat resistance clothing).
- Remove immediately flammable matter.
- Use specified extinguisher or extinguishing equipments.
- Water to cool, when sealed container is exposed to high temperature.
- Distinguish to leeward.

## 6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures
- Wear proper protectors (gloves, respirator, safety goggles, apron, etc.).
- Prohibit the outskirts from an entrance and don't bring it close anyone other than the person concerned and prevent a second disaster.
- Remove immediately ignition source or heat source and the other flammable matters.
- Furnish with extinguisher proper for accidental fire.
- Avoid shock or heating.

Ecological information
- Take precautions so as not to pollute rivers, lakes, etc.

Methods and materials for containment and clean up
- Recollect leak in container and keep in safety place with the container closed.
- Dispose of waste or clinging matter according to related regulations.

- Recollect with tool of non-spark material.
- Recollect by absorbing with dry sand or soil or other non-flammable matter. Prevent from flowing out by banking, if in large volume.

## 7. Handling and Storage

**Handling**

- Handle in well ventilated area and close tightly container with closer necessarily during and after use.
- Close tightly container with closer necessarily during and after use.
- Take away from open flame, spark or heat-source.
- Wear antistatic suits and shoes, while working.
- Use handtool of spark proof type.
- Ground equipment for the prevention of static electricity accident because of generating easily static electricity and use electric appliance or apparatus of explosion proof type because of containing organic solvent.
- Avoid shock and heating.
- Avoid mixing with foreigen matter and contact with acid, amine, metal, and combustible matters (paper, wood, etc.).
- Keep paint sludge, paint dust and rag or paper or roller soiled by paint soaking in water before dumping. These may spontaneously ignit, if piled up.
- Wear proper protectors to avoid contact of skin, eyes and mucous and, of clothing.
- Wash thoroughly hand and face after handling. Avoid bringing contaminated protectors in any place where person stays.
- Install proper local ventilator and wear proper protector in closed space.
- Do not let a person who has ever suffered allergy handle.
- Perform operations including application according to "Foul Smell Control Law" because paint products and materials and their waste are specified as a foul smell material to be regulated in the law.

**Storage**

- Avoid direct sunlight.
- Store in good ventilation.
- Store with keeping away from ignition or heat source.
- Store in cool, dry, dark place.
- Prevent from overturning and falling.
- Avoid storing together with other chemicals. Avoid keeping especially with acid, amine, metal and combustible matters (wood, paper, fabric, etc.).

## 8. Exposure Controls and Personal Protection

| No. | Components Name | C.C | ACGIH(TLV-TWA) |
|---|---|---|---|
| 1) | Amorphous silica A | | |
| 2) | Silane, dichlorodimethyl, reaction product | | 10M |
| 3) | Titanium Dioxide | | |
| 4) | Iron hydroxide oxide | | |
| 5) | Xylene | 50P | 100P |
| 6) | Ethyl Benzene | 20P | 100P |
| 7) | Methanol | 200P | 200P |
| 8) | Isobutyl Alcohol | 50P | 50P |
| 9) | Formaldehyde | 0.1P | |
| 10) | Oxidative-polymerizable Resin | | |
| 11) | C.I.Pigment green 7 | | |

*Sign : P:ppm M:mg/m3 C.C:Control Criteria SD:Skin damage CR:Corrosive FT:Fish Toxicity

**Equipment**

- Use explosion proof type equipment.
- Install proper exhauster to prevent concentrating of vapor.
- Ground equipment for fluid transfer, pumping and stirring.
- Use equipment devised so that heat or ignition source can not be placed in the neighborhood of handling place.
- In indoor application, use automatic applicator or install proper equipment or local exhauster to avoid direct exposure to spray mist.
- In application in closed space or the like (tanks), install equipment which can sufficiently ventilate even its bottom.
- In handling for long time, use equipment devised so that sufficient

ventilation can be ensured and exposure to exhaust gas can be avoided.

Respiration and Ventilation
- Wear gas mask for organic gas.
- Wear air mask in closed space.

Hand protection
- Wear gloves impervious to organic solvent and chemical.

Eye protection
- Wear safety goggles.

Skin and body protection
- Wear suitable clothing to avoid direct contact of skin when handle. The clothing should be chemical-proof.

Other
- Wear electricity conductive shoes in electrostatic application.

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## 9. Physical and Chemical Properties

Physical state

| | | | |
|---|---|---|---|
| Appearance | : Liquid | | |
| Color | : Specified color | | |
| Odor | : Organic solvent smell | | |
| pH. | : N.AV. | Melting point/freezing point | : N.AV. |
| Boiling Point | : 107.9 - 141 Degrees C | Flash Point | : 22.5 Degrees C |
| Explosion Limit | : (lower) 1.1 % | (upper) 10.9 % | |
| Vapor Pressure | : 1333/Pa | | |
| Relative density | : N.AV. | Solubility | : N.AV. |
| Density | : 1.17g/cm3 | | |
| Partition coefficient:n-octanol/water | : N.AV. | | |
| Ignition Point | : 430 Degrees C | Decomposition temperature | : N.AV. |
| Odour Threshold | : N.AV. | Flammability(solid,gas) | : N.AV. |
| Evaporation rate | : N.AV. | | |
| Other | : | | |

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## 10. Stability and Reactivity

Stability

Additional Information
- Does not react under normal conditions.

Conditions to avoid
- N.AV.

Incompatible materials
- N.AV.

Hazardous decomposition products
- Emitts hazardous gases of carbon monoxide and low molecular weight monomeres.
- Emitts hazardous gases of carbon monoxide and nitrogen oxides.

Other hazardous information
- May explode when burnt in large volume.
- May spontaneously ignit when the resin component oxidizes in the air, generates heat or stores heat.
- Avoid leaving cloth, paper, brush or roller which contains the paint, or its dust to accumulate or to be made round.

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## 11. Toxicological Information

| No. | Components Name | LD50M | LD50S | Sk | Ey | Br | Mu | Ca | Re | Si | | Rp | As | EtC |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1) | Amorphous silica A | | | | | | | | | | | | | |
| 2) | Silane, dichlorodimethyl, reaction product | | | | | | | | | | | | | |
| 3) | Titanium Dioxide | 20000 | 10000 | | 2B | | | | | | | | | |
| 4) | Iron hydroxide oxide | | | | | | | | | | | | | |
| 5) | Xylene | 3500 | | 2 | 2A | | | | 1B | 1 | | 3 | 1 | 2 |
| 6) | Ethyl Benzene | 3500 | 15400 | 3 | 2B | | | 2 | 1B | | 2 | 3 | | 1 |
| 7) | Methanol | 1400 | 15800 | | 2 | | | | 1B | 1 | | 3 | 1 | |
| 8) | Isobutyl Alcohol | 2596 | 2523 | 2 | 2A | | | | | | | 3 | | 2 |
| 9) | Formaldehyde | 605 | 270 | 2 | 2A | 1 | 2 | 1A | | 1 | | | 1 | |
| 10) | Oxidative-polymerizable Resin | | | | | | | | | | | | | |
| 11) | C.I.Pigment green 7 | | | | | | | | | | | | | |

Sign :LD50M :mouth(oral-rat) mg/kg LD50S:skin(rabbit) mg/kg
Sk :Skin corrosion/irritation Ey :Serious eye damage/eye irritation
Br :Respiratory sensitization/Skin sensitization
Mu :Germ cell mutagenicity Ca :Carcinogenicity Re :Reproductive toxicity
Si :Specific target organ systemic toxicity-Single exposure
Rp :Specific target organ systemic toxicity-Repeated exposure
As :Aspiration hazard

Other hazardous information
- May cause rash or eruption in contact with skin.

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## 12. Ecological Information

- Take cautions in handling because it may contaminate environment

when it leaks or spills or it is dumped. Especially, take cautions for product or rinsing water not to flow into the ground, river or ditch.

| | |
|---|---|
| Ecotoxicity | N. AV. |
| Persistence and degradability | N. AV. |
| Bioaccumulative potential | N. AV. |
| Mobility in soil | N. AV. |

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## 13. Disposal Considerations

**Residual waste**

- Let authorized industrial waste handling agent dispose of paint waste, container waste, etc. by making service agreement.
- Do not dispose of waste water after cleaning container, equipment, tool, etc. into the ground and drainage ditch without treatment.
- Dispose of waste from effluent treatment, incineration, etc., compling to regulations or acts relating to waste treatment or cleaning or let special handling agent dispose of them.
- Burn up paint waste, additive waste, etc. little by little using incineration of open type with them absorbed in diatomaceous earth.
- Let authorized industial waste handling agent by making service agreement, because these are specified as Special Control Industrial Waste(waste oil).
- Dispose of paint product and material and their waste according to "Foul Smell Control Law" because these are specified as a foul smell material in the law.

**Contaminated container and packing**

- Dispose of used container after emptied completely.
- Let authorized industial waste handling agent by making service agreement.

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## 14. Transport Information

**General**

- Accord to the description in "Precautions in Handling " and "Precautions in Storing." Make sure that container does not leak and be sure to prevent it from collapsing by loading with it so that it may not overturn or fall down or sustain damage.

**Domestic regulation**

**Land Transportation**

- Accord to each transportation method specified in"Fire Service Law", "Industrial Safety and Health Law"or "Poisonous and Deleterious Substances Control Law", when applicable to any of them.
- The load sending person the conveyance note book ( Yellow card etc.) delivers to the transporter.

**Maritime Transportation**

- Accord to requirements specified in "the Vessels Safety Law (the Ship Safety Act").

**Air Transportation**

- Accord to requirements specified in "the Aviation Act."

**Other**

**International regulation**

| | |
|---|---|
| U.N. No. | 1263 |
| UN transporter shipping) name | Paint or paint related material |
| IMDG | 3 |
| Packing group | II |

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## 15. Regulatory Information

"Fire Service Law": Dangerous Materials Class 4 Flammable Liquids Group 2 Petroleums.
"Enforcement Order of Industrial Safety and Health Law" : Flammable Subtances
"Regulation on Organic Solvents Poisoning Prevention" : Article 1, Clause 1, Item 2 (Class 2 Organic Solvents)
"Industrial Safety and Health Law": Article 57 (Harmful Materials Which Shall Be Labeled)
"Industrial Safety and Health Law": Article 57-2
"Preventive Regulation on Particularly Specified Chemical Substances Trouble": Article 2, Clause1, Item3-3(Special Organic Solvents, containing more than 1wt% either ethylbenzene, styrene, and methyl isobutyl ketone)
"Foul Smell Control Act"
"Law concerning Reporting, etc. of Releases to the Environment of Specific

| | |
|---|---|
| | Chemical Substances and Promoting Improvements in Their Management ″ : Particularly Class 1 Specified Chemical Substances.<br>″Law concerning Reporting, etc. of Releases to the Environment of Specific Chemical Substances and Promoting Improvements in Their Management ″ : Class 1 Specified Chemical Substances. |
| 16. Other Information<br>Literatures | · ″Raw Material Data Base″, edited by Japan Paint Manufacturers Association<br>· ″GHS Compliant SDS / Labelling Guidebook for mixtures (paints)″ Japan Paint Manufacturers Association<br>· SAX's″Dangerous Properties of Industial Materials″, Van Nostrand Reihold, New York. |
| Remarks | · This ″SDS″ is based on relevant information which has been available. The information or assessment on safety of the product is not necessarily satisfactory for the reason that it is assessed from that of relevant individual materials. Handle safely the product or material on your responsibility according to this ″SDS″.<br>· This ″SDS″is subject to change without notice.<br>· The content, physical datum or chemical datum of the product in the ″SDS″ is to be an indicator for the relevant product quality or it has not anything decisive to do with the quality.<br>· This safety information stems from standards of Japan Paint Manufacturers Association with laws of the government but not from laws of local governments. When handled, exhausted and wasted, accord to relevant laws of local governments as required. |